Project 2

Firoz’s Pizza Express

Context

Firoz is a young entrepreneur from Mumbai who grew up around street food stalls in Mohammad Ali Road. He always dreamt of combining India's love for food with the convenience of modern tech.

In 2024, he launched ***Firoz's Pizza Express***, an on-demand pizza delivery service. Unlike Domino's or Pizza Hut, Firoz wanted to build something lean, run mostly through WhatsApp orders and freelance delivery partners (just like Zomato's delivery model).

He started small: only two kinds of pizzas at the beginning –

- **Margherita (₹400)** – simple cheese, tomato, and basil.
- **Paneer Tikka (₹600)** – with spiced paneer and capsicum, to give an Indian twist.

Customers could also customize their pizzas with extra toppings like olives, capsicum, or extra cheese (₹50 each), or ask for certain toppings to be excluded.

To deliver pizzas, Firoz recruited a small team of runners—college students and gig workers—who registered via Google Forms and picked up deliveries on a per-order basis.

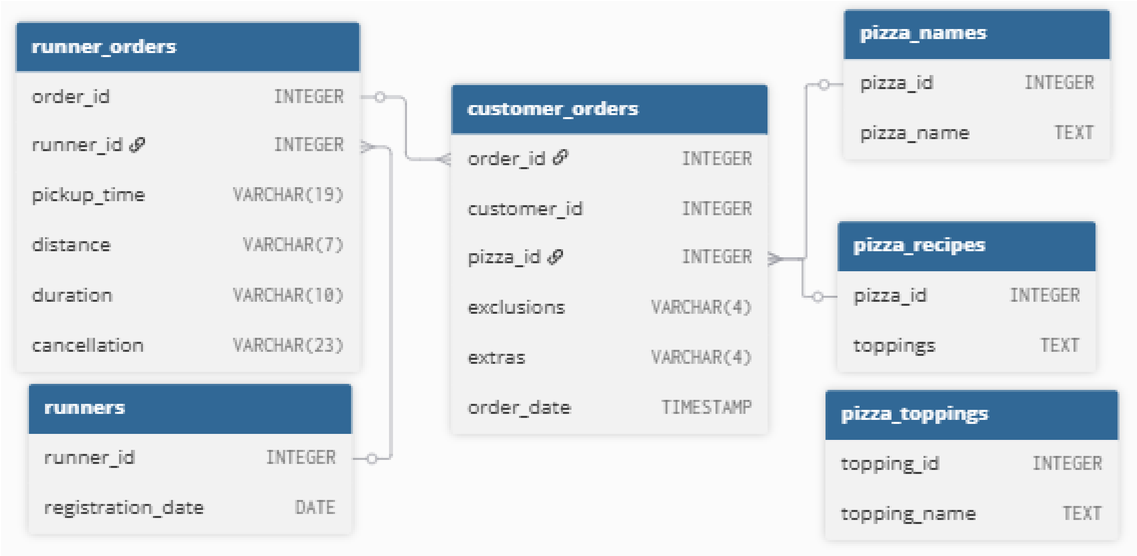
He started capturing data in Excel, but as the business grew, he realized that data-driven decisions were the only way to scale. He asked you and your team (the students!) to help him clean the data, analyze it, and extract insights.

Dataset Overview

Firoz has been recording the following:

- Runner registrations – when each runner signed up.
- Customer orders – each row represents one pizza in an order.
- Runner assignments – which runner picked up which order, and whether it was delivered or cancelled.
- Pizza catalog & recipes – the standard pizzas and their ingredients.
- Ingredient reference – a mapping of ingredient IDs to ingredient names.

Entity Relationship Diagram



Sample Data

1. runners - The runners table shows the *registration_date* for each new runner.

runner_id	registration_date
1	2025-06-01
2	2025-06-05
3	2025-06-10
4	2025-06-15

2. customer_orders - Customer pizza orders are captured in the *customer_orders* table with 1 row for each individual pizza that is part of the order.

   The *pizza_id* relates to the type of pizza which was ordered whilst the exclusions are the *ingredient_id* values which should be removed from the pizza and the extras are the *ingredient_id* values which need to be added to the pizza.

   Note that customers can order multiple pizzas in a single order with varying *exclusions* and *extras* values even if the pizza is the same type!

   The *exclusions* and *extras* columns will need to be cleaned up before using them in your queries.

order_id	customer_id	pizza_id	exclusions	extras	order_time
1	201	1			2025-06-20 18:00:00
2	202	1			2025-06-20 18:05:00
3	203	2		5	2025-06-21 19:30:00
3	203	1	4		2025-06-21 19:30:00
4	204	2			2025-06-22 20:00:00
5	205	1	4	5	2025-06-22 21:15:00
6	202	2		6	2025-06-23 17:45:00
7	206	1			2025-06-23 18:30:00
8	207	2		1,6	2025-06-23 19:15:00

3. runner_orders - After each order is received through the system - they are assigned to a runner - however not all orders are fully completed and can be cancelled by the restaurant or the customer.

The *pickup_time* is the timestamp at which the runner arrives at the Pizza Express headquarters to pick up the freshly cooked pizzas. The *distance* and *duration* fields are related to how far and long the runner had to travel to deliver the order to the respective customer.

order_id	runner_id	pickup_time	distance_km	duration_mins	cancellation
1	1	2025-06-20 18:10:00	5.2	15	
2	2	2025-06-20 18:15:00	3.8	12	
3	1	2025-06-21 19:40:00	7.5	20	
4	3	NULL	NULL	NULL	Restaurant cancelled
5	4	2025-06-22 21:25:00	4.5	14	
6	2	2025-06-23 17:55:00	6	18	
7	3	2025-06-23 18:40:00	2.5	9	
8	1	2025-06-23 19:25:00	8.2	22	

4. pizza_names - At the moment - Pizza Runner only has 2 pizzas.

pizza_id	pizza_name
1	Margherita
2	Paneer Tikka

5. pizza_recipes - Each *pizza_id* has a standard set of ingredients which are used as part of the pizza recipe.

pizza_id	ingredient_id
1	1 (Cheese)
1	2 (Tomato)
1	3 (Basil)
2	1 (Cheese)
2	4 (Paneer)
2	5 (Capsicum)

6. ingredients - This table contains all of the *ingredient_name* values with their corresponding *ingredient_id* value.

ingredient_id	ingredient_name
1	Cheese
2	Tomato
3	Basil
4	Paneer
5	Capsicum
6	Olives

Goal

By the end of this project, you will gain hands-on practice with joins, aggregation, date/time manipulation, string aggregation, conditional logic, CTEs and window functions — all while analyzing a real-world inspired pizza express business case.

Analytical Questions

A. Runner Activity Metrics

1. How many runners signed up each week? (Assume week starts Monday.)
2. What's the average time (in minutes) between runner registration and their first pickup?
3. What's the successful delivery count per runner?
4. Delivery success rate per runner (ratio of non-cancelled to total assigned orders)?
5. Average speed per runner (distance/duration); any noticeable trends?

B. Order & Delivery Insights

1. Total number of pizzas ordered.
2. Unique customer orders count.
3. Average waiting time (order_time to pickup_time) for successful deliveries.
4. Difference between longest and shortest delivery durations.

C. Ingredient Optimization

1. List standard ingredients per pizza (by name).
2. Most commonly added extra ingredient (name).
3. Most commonly excluded ingredient.
4. Total count of each ingredient used in delivered pizzas, sorted descending.